

Yen-Hsiang (Robert) Huang

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Education

Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0

Georgia Tech, Atlanta, GA, USA

Sept. 2022 – Dec. 2027 (expected)

B.S. in Electrical Engineering | GPA: 4.06/4.30

National Taiwan University (NTU), Taipei, Taiwan

Sept. 2017 – July 2022

Research Experience

Graduate Research Assistant

SCCAD (former GTCAD), Georgia Tech, Advisor: Prof. Sung Kyu Lim

Sept. 2022 – Present

3D IC Optimization

Fall 2024 – Present

- Timing optimization through reduced critical-path snaking and inter-die cell relocation (Snake-3D).
- Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs, incorporating interconnect RC effects into timing-aware clock-tree optimization (CLK-3D).

Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES

Fall 2023 – Spring 2024

- Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs.
- Reduced WNS of legacy-node 3D ICs by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.

Die Bonding Bumps and Pads Legalization for 3D ICs

Sept. 2022 – May 2023

- Eliminated all 3D-via overlaps when via utilization is less than 40%.
- Developed 3D-via legalization with refinement and minimal runtime overhead (papers: ISPD, TCAD).

Individual Study

EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang

Apr. 2020 – Sept. 2021

Independent Study on Physical Design

- Participated in physical-design-related global contests three times across different teams and placed in the top five twice (ISPD contest 2021 and ICCAD contest 2021).

Individual Study

ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang

Sept. 2020 – June 2021

Independent Study on Reversible Circuit Synthesis

- Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

Work Experience

R&D Engineer Intern

Synopsys Inc.

Sunnyvale, CA

May 2025 – Dec. 2025

- Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.
- Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.

Software Engineer R&D Intern

Synopsys Inc. – Concurrent Clock-and-Data (CCD) Team

Hsinchu, Taiwan

July 2021 – Aug. 2021

- Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering flip-flop timing (slack).
- Improved timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack.

Software Engineer R&D Intern

Cadence Inc. – Conformal Team

Hsinchu, Taiwan

June 2020 – Sept. 2020

- Provided engineering change order (ECO) test cases in a five-person team for Cadence Conformal's robustness test.
- Generated 2,000+ cases across five categories in which Conformal acted unusually and proposed three improved flows to address them.

Publications

- Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025.
- Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," ACM International Symposium on Physical Design (ISPD), 2023.

Selected Projects

Mixed-Size Macro Placement and Legalization

Feb. 2021 – July 2021

- Developed a mixed-size macro placer for non-rectangular designs within a two-person team, maximizing the area for standard cells while maintaining the minimum channel needed by macros.

Honors and Awards

Ministry of Science and Technology Undergraduate Research Grant

July 2021

- Topic: Wafer-Scale Physics Modeling for Finite Element Methods.

Honorable Mention Award in the ACM ISPD Contest 2021

March 2021

- 4th place worldwide in the contest.

TSMC Scholarship for Undergraduate Students

Sept. 2020 – Jan. 2021

- Awarded to approximately 20 undergraduate students in Taiwan every semester.

The Dean's List Award

Spring 2020

- Ranked 1st, with a 4.30/4.30 GPA in the semester.

4th in Data Structure and Programming Contest

March 2019

- Cadence Inc. sponsored the contest, with 125 contestants.

Skills

Programming Languages

- C/C++ (expert), Python, MATLAB, $\text{\LaTeX}$.

Languages

- English, Mandarin (native).